



Assessment of the oxidative and genotoxic effects of the glyphosate-based herbicide roundup on the freshwater shrimp, *Macrobrachium nipponensis*

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HIGHLIGHTS

- An acute toxic test was conducted to evaluate the effect of Roundup on oxidative stress and genotoxicity from *Macrobrachium nipponensis*.
- Roundup could induce significant oxidative stress and genotoxic effect by sublethal concentration exposure.
- All biomarkers except CAT were considered sensitive for toxicity assessment of Roundup exposure.

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ABSTRACT

In the present study, an acute toxic test was performed to assess the oxidative stress and genotoxic effects of the herbicide on the freshwater shrimp *Macrobrachium nipponensis*. The results showed that the 48-h and 96-h LC₅₀ values of Roundup to *M. nipponensis* were 57.684 mg/L and 11.237 mg/L, respectively. For further investigation, the shrimps were exposed to sublethal concentrations of 0.35, 0.70, 1.40, 2.80 and 5.60 mg/L for 96 h. A significant decrease in total haemocytes count (THC) was observed at concentration of 5.60 mg/L throughout the experiment. The level of superoxide dismutase (SOD), catalase (CAT) and total antioxidant capacity (T-AOC) in all the treatments decreased in a dose- and time-dependent manner except for the concentration group of 0.35 mg/L. The malondialdehyde (MDA), hydrogen peroxide (H₂O₂) and protein carbonyl in serum increased significantly at concentrations of 2.80 mg/L and 5.60 mg/L. A significant decrease in acetylcholinesterase (AChE) activity was observed at each concentration ($P < 0.05$). In addition, the micronucleus (MN) frequency of haemocytes significantly increased ($P < 0.05$) at concentrations of 1.40, 2.80 and 5.60 mg/L, whereas the comet ratio and %DNA in the tails exhibited a clear time- and dose-dependent response during the exposure. The analysis of the integrated biomarker response (IBR) showed the induction of oxidative stress biomarkers and the inhibition of antioxidants, and this dose-dependent relation suggests the sensitivity and availability of all the biomarkers. These results revealed that Roundup had a prominent toxic effect on *M. nipponensis* based on the antioxidative response inhibition and genotoxicity.

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1. Introduction

As the best-selling herbicides worldwide, glyphosates are widely utilised in both agricultural and non-agricultural areas, such

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as aquatic waters, for weed control (Myers et al., 2016). The major role of glyphosate is the inhibitor of the enzyme 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS) in the shikimate acid pathway. Because of the absence of the shikimate pathway in non-target organisms, glyphosate is considered to have low toxicity to animals (Felder, 2005). The glyphosate-based commercial formulation Roundup (RD) was developed by Monsanto Co. (St. Louis, MO, USA) in 1970. The use of RD rapidly increased in the recent two decades in China, owing to the massive growth of crop cultivation and the introduction of transgenic plants

resistant to glyphosate. Glyphosate is slightly toxic to aquatic animals according to data from the World Health Organization (WHO, 1994); however, RD is considered more toxic probably due to the surfactant polyoxyethylene amine (POEA) (Brausch and Smith, 2007). The extensive use of RD on crop fields may lead to environmental hazards in aquatic organisms typically living in areas close to cultivars, especially in aquacultural ponds where Roundup is directly used to control aquatic plants (Vick, 2010). In recent decades, several studies have demonstrated that glyphosate, or glyphosate-based herbicide, is toxic to non-target aquatic organisms, including microorganisms (Arunakumara et al., 2013; Wu et al., 2016), aquatic plants (Zhong et al., 2017), invertebrates (Mensah et al., 2012), amphibians (Edge et al., 2013) and fishes (Bridi et al., 2017; Roy et al., 2016; Samanta et al., 2014; Uren Webster and Santos, 2015). However, most studies have focused on fishes, and no reports regarding economical crustaceans, such as shrimps, has been published.

The freshwater shrimp, *Macrobrachium nipponense*, is the main species widely bred in China with an annual output of more than 270,000 t. Rice-shrimp co-culture is one of the most important and widespread modes of breeding in China, particularly in Jiangsu Province (BFMA, 2017). The wide application of glyphosate in China is becoming a potential threat to shrimps bred in rice-shrimp coculture systems. Because of its relatively long half-life in water (most commonly 45–60 days) (Annett et al., 2014) and repeated application in practice, the residue of glyphosate in aquatic waters could reach up to 0.765 mg/L 24 h after spraying in a breeding pond (Zhu et al., 2000). As bottom dwellers, shrimps are more frequently in contact with xenobiotics in water and sediment, making these organisms more at risk to deterioration in water quality, such as the directly exposure of pesticides. Therefore, it was necessary and well worth to investigate the potential toxicity of glyphosate-base herbicides on *M. nipponense* for the safe assessment of aquaculture and the aquatic ecological environment.

Pesticides, including herbicides cause oxidative stress to aquatic animals by inducing the excessive production of reactive oxygen species (ROS), including hydrogen peroxide (H_2O_2), superoxide ion ($O_2^{\cdot-}$) and hydroxyl radical ($\cdot OH$). ROS are highly reactive substances that cause damage to lipids, proteins, carbohydrates and nucleic acids (Lushchak, 2011). Animals have developed an antioxidant system to equilibrate the antioxidant defence and the generation of ROS. In this system, enzymatic antioxidants, such as superoxide dismutase (SOD) and catalase (CAT), serve as the first line of defence. Thus, the SOD and CAT activities in different tissues were proposed as indicators of pollutant-mediated oxidative stress (Li et al., 2003; Samanta et al., 2014). Numerous studies have demonstrated that organophosphate (OP) pesticides can cause genotoxic damage by mutations and chromosomal aberrations in aquatic animals. As a sensitive biomarker of chromosome damage, the micronucleus (MN) assay has been applied in studies with various freshwater and marine vertebrates and invertebrates to assess the genotoxicity of pollutants, such as heavy metals and pesticides (Barka et al., 2016; Bolognesi and Hayashi, 2011; Bolognesi et al., 2006). Additionally, the comet assay, which performed by a single cell gel electrophoresis, is widely used to assess DNA damage from environmental pollution in aquatic animals (Lee and Steinert, 2003). However, few studies have reported the evaluation of the genotoxic effect of glyphosate-based herbicides on freshwater shrimps by the MN and comet assays. In this sense, the simultaneous application of the MN and comet assays to investigate the genotoxicity of RD would be a valuable contribution.

In the present study, the THC, the level of antioxidants, including SOD and CAT, the total antioxidant capacity (T-AOC), the oxidative damage products, including malondialdehyde (MDA), hydrogen peroxide (H_2O_2) and carbonyl protein (PC), and the MN and comet

assays were measured to assess the oxidative stress and genotoxicity of the glyphosate-based herbicide, Roundup, to *M. nipponense*. Moreover, acetylcholinesterase (AChE) activity was also detected in this context as a sensitive biomarker in organophosphate pesticide exposure (Monserrat et al., 2002).

2. Methods and materials

2.1. Animals and maintenance

The experiment was conducted in November 2017 at Xichang University, Xichang, China. All the procedures were authorised by the Animal Bioethics Committee of Xichang University. Adult freshwater shrimp *M. nipponense* (Crustacean: Decapoda: Macrobrachium), with an average weight of 4.42 ± 0.76 g, were collected from Qionghai Lake in Xichang. The shrimps were acclimated to laboratory conditions for 14 days prior to the start of the experiment. Eighteen $150\text{ cm} \times 100\text{ cm} \times 120\text{ cm}$ glass aquaria containing filtered freshwater were utilised in this period. In addition, each four ultraviolet-treated PVC tubes were placed in an aquarium as shelter. The aquacultural system included regulated photoperiod (12 h:12 h) and constant temperature ($20\text{--}22^\circ\text{C}$). The animals were fed at 7:00 p.m. once a day with commercial shrimp ration, and residual food debris and faeces were removed 2 h after feeding. The following water conditions were monitored daily: temperature $20 \pm 1.0^\circ\text{C}$, pH 7.3 ± 0.5 , dissolved oxygen 7.6 ± 0.3 , ammonia nitrogen $< 0.2\text{ mg/L}$, and nitrite $< 0.005\text{ mg/L}$. Roundup (41% glyphosate, Sinochem Crop Care Co. LTD, Shanghai, China) used in this test was purchased from the local market.

2.2. Median lethal concentration (LC_{50}) of roundup on *M. nipponense*

The animals were starved for 2 days prior to each experiment to mitigate potential stress. Acute static toxicity tests were conducted by exposing the animals to different concentrations of Roundup, following a pre-experiment test that determined the maximum and minimum concentration. Roundup was dissolved in water from the aquarium to concentrations of 10, 18.2, 33.1, 60.3, 109.7 and 200 mg/L. Water without any Roundup served as the control. Each treatment included 3 replicates with 30 shrimps in each treatment, totalling 210 shrimps that were used in this experiment. Healthy shrimps without disability were randomly selected and allocated in the glass aquarium with 100 L of freshwater and 10 shrimps per container. Then, the 48-h and 96-h median lethal concentrations of Roundup for *M. nipponense* were determined. Water conditions were not changed and no feeding, aeration or mixing throughout the experiment. Status of animals were checked every 2 h and shrimps with no movement of the appendages in response to a stick poke were confirmed as dead and immediately removed. The LC_{50} values and 95% confidence limits were calculated by the probit regression analysis with the SPSS V20.0 statistical software.

2.3. Sublethal concentration exposure

Considering the results from J. Vera-Candioti (Vera-Candioti et al., 2013), who reported that there is no significant difference between the measured glyphosate concentration and 24-h interval renewals of the testing solutions, and that there was a less than 15% reduction of glyphosate in water after 96 h exposure (Sinhorin et al., 2014), the level of glyphosate in the water was not measured for both sublethal tests and acute toxicity tests. Based on the 48-h and 96-h LC_{50} values and the safe concentrations of Roundup on *M. nipponense*, five nominal concentrations (0.35, 0.70, 1.40, 2.80 and 5.60 mg/L), representing approximately 1/32, 1/16, 1/

8, 1/4 and 1/2 of the 96-h LC₅₀ values, respectively, as well as a control group that received no Roundup, were established in this experiment. Each treatment was replicated three times with 40 shrimps in each replicate, and resulting in a total of 720 shrimps being used. The characteristics of the aquarium water were the same as those in the acute toxicity test. The water in the aquaria was half-changed daily by adding fresh water containing the same concentrations of Roundup to ensure the invariance of concentration in each group during the experiment.

2.4. Sampling

Following exposure, at least nine animals were randomly removed from three independent aquaria for each treatment at 0, 24, 48, 72 and 96 h. The shrimps were immediately anaesthetised in an ice bath, and haemolymph was withdrawn from the joint of the carapace and sternite with a sterile syringe. To ensure a sufficient quantity of sample for each analysis, haemolymph from three individuals was pooled as one replicate, and three replicates with totally 9 shrimps were randomly taken for the assays at each time point. Some haemolymph was immediately 1:1 diluted with anticoagulant (30 mM trisodium citrate, 338 mM NaCl, 115 mM glucose, and 10 mM EDTA) for the MN assay, and some was centrifuged at 1200 r/min for 5 min to collect the haemocytes for the THC and comet assay. The remainder of the haemolymph sample was stored at 4 °C for the subsequent assays of other parameters.

2.5. Total haemocytes count

The THC was performed by a haemocytometer under a Motic AE21 inverted phase-contrast microscope (Motic, Canada). Three replicates were made for each sample, and mean value was recorded for statistical analysis. The THC was represented as haemocyte numbers per milliliter haemolymph.

2.6. Antioxidant activity

Haemolymph was pre-cooled in a refrigerator (4 °C) and centrifuged at 11,000 r/min at 4 °C for 30 min, and the supernatant was carefully collected as serum. Then, the serum from each replicate was placed into sterile centrifuge tubes and stored at 4 °C for use in the oxidative parameter assays described below. The antioxidant, SOD and CAT activities in serum were spectrophotometrically determined at 550 nm and 405 nm with the corresponding detection kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's protocols. One unit of SOD was defined as the enzyme activity that inhibited the photoreduction of NBT to blue formazan by 50% and was expressed as U/mL. One unit of CAT activity was defined as the amount of enzyme that catalysed the decomposition of 1 mmol of H₂O₂ per min and was expressed as U/mL. T-AOC was measured at 593 nm with the ferric reducing ability of plasma (FRAP) assay (Benzie and Strain, 1996) with a detection kit, according to the manufacturer's protocols (Nanjing Jiancheng Bioengineering Institute, China) and was also expressed as U/mL.

2.7. Oxidative damage parameters

Oxidative damage was represented as lipid peroxidation products (measured as MDA), reactive oxygen species (measured as H₂O₂), and protein damage (measured as protein carbonyl). MDA and H₂O₂ in serum were spectrophotometrically detected at 532 nm and 405 nm with the corresponding detection kits (Nanjing Jiancheng Bioengineering Institute, China) according to the manufacturer's protocols. The MDA and H₂O₂ levels were expressed as

nmol/L and mmol/L, respectively. For the protein carbonyl assay, an ultraviolet spectrophotometric method was performed at 370 nm with a detection kit (Nanjing Jiancheng Bioengineering Institute, China). The protein carbonyl level in serum was expressed as nmol carbonyl per mg of protein⁻¹. The protein content in the samples (data not shown) was spectrophotometrically determined as previously described (Lowry, 1951) using bovine serum albumin as a standard.

2.8. Acetylcholinesterase

The acetylcholinesterase enzyme activity was measured by the spectrophotometric method with a detection kits following the manufacturer's protocol (Nanjing Jiancheng Bioengineering Institute, China). The absorbance was determined on a microplate reader (iMark, Bio-Rad, USA) at 415 nm, and the activity of enzyme was expressed as U/mL.

2.9. Kinetics assay

The kinetics assay was conducted to investigate the inhibitory activities of Roundup to SOD, CAT and AChE. SOD activity was detected spectrophotometrically and the ODs were recorded every 30 s during the 20 min reaction as introduced above. Different volume of 750 μM NBT (0.05, 0.1, 0.2, 0.4, 0.8 mL) as indirect substrate was added to the reaction system respectively. CAT activity was determined under an ultraviolet spectroscopy method at 240 nm and different concentration of H₂O₂ were added to the reaction system with 1 mM, 5 mM, 10 mM, 15 mM and 20 mM respectively. The activity of AChE was measured according to the method by Gao et al. (1996) that different concentration (0.2 μM, 0.4 μM, 0.6 μM, 0.8 μM and 1.0 μM) of acetylthiocholine iodide (ATCh) was mixed with enzymes as substrate and incubated at 30 °C for 5 min. 1.8 mL of dinitrobenzene was added to terminate the reaction. ODs were recorded at 415 nm. The Michaelis constant (K_m) for each enzyme was calculated by the Lineweaver-Burk Plot. To analyse the inhibitory effect of Roundup to each enzyme, 0.5 mM of NBT, 50 mM of H₂O₂ and 2 μM of ATCh were used as substrates and Roundup concentrations used in the assays were fixed at 0, 0.625, 1.25, 2.5, 5.0 and 10.0 mg/L, respectively. Enzyme-inhibitory activity was calculated by the following equation:

$$\text{Inhibitory activity(\%)} = \left[\frac{(\Delta A/\text{min})_{\text{control}} - (\Delta A/\text{min})_{\text{sample}}}{(\Delta A/\text{min})_{\text{control}}} \right] \times 100\%$$

Where (ΔA/min)_{control} and (ΔA/min)_{sample} were the enzyme activity in the absence and presence of Roundup, respectively. ΔA/min values were calculated by subtracting the absorbance every minutes. The IC₅₀ value was estimated by nonlinear regression using GraphPad Prism 5.0 software. In addition, several concentrations of AChE (0.1, 0.2, 0.4, 0.8 μg/mL) were added to the reaction system after the addition of Roundup and the residual activities were determined. The inhibitory type of Roundup was analysed based on the plot with enzyme concentrations against activities.

2.10. MN (micronucleus) assay

The MN assay was performed according to the method from (Barka et al., 2016), with slight modifications. The haemocytes collected in section 2.4 were fixed (1% glutaraldehyde and 1% sucrose) for 30 min at 4 °C and then dyed with Gimesa solution (Sigma, Germany) for 5 min and rinsed with distilled water 3 times. The slides were air-dried and observed under an inverted microscope (AE21, Motic, Canada) with 10 × 100 fold magnification. At

least one thousand cells were scored per slide to determine the MN frequency. An MN was defined as a round structure with a diameter of less than 1/3 of that of the primary nucleus but on the same optical plane as that of the primary nucleus, and the boundaries of the MN could be distinguished from those of the primary nucleus (Barka et al., 2016). Three slides were observed for each replicate, and the MN frequency was expressed as the number of cells containing micronucleus/total cells.

2.11. Comet assay

The alkaline comet assay was adapted from a previously described method (Singh et al., 1988). The first layer was made by placing 100 μ L of 0.8% normal melting agarose (NMA) on a frosted microscope slide and covered with a coverslip immediately. Slides were stored at 4 °C for 10 min to stabilise the layer. After haemocytes collection, 50 μ L of 1% low melting agarose (LMA) was mixed with 25 μ L of cell suspension with cell density of 1×10^6 ml⁻¹, and layered over the first agarose layer to make the second layer. Slides were also covered with a clean coverslip and stored at 4 °C for 10 min. Then, 75 μ L of 0.8% LMA was carefully added on the layers after removal of the coverslip and cooled at 4 °C for 30 min. Subsequently, the slides were immersed in a cell lysing solution (2.5 mol/L NaCl, 10 mmol/L Tris–HCl, 100 mmol/L Na₂EDTA, 10% DMSO and 1% Triton X-100, pH 10) at 4 °C for 40 min. Slides were rinsed twice in cool phosphate buffer solution (PBS) and placed in a horizontal electrophoresis chamber filled with alkaline electrophoresis buffer (300 mmol/L NaOH, 1 mmol/L EDTA, pH 13) for 20 min to promote DNA unwinding. Electrophoresis was carried out under 25 V and 300 mA for 20 min, and the slides were carefully removed and neutralised for 5 min in a neutralisation buffer (0.4 M Tris–HCl, pH 7.5) for three times. Finally, the slides were stained with 20 μ L of 0.1-mg/ml propidium iodide (PI) for 5 min and observed under an inverted fluorescence microscope (DM-i8, Leica, Germany). For each treatment, three slides with haemocytes from nine shrimps, and ten random fields with at least 100 cells from each slide were analysed by the Comet Assay Software Project (CASP). For DNA damage evaluation, two commonly used parameters, the comet ratio (cells with tail/all cells) and the percentage of DNA in the comet tails were calculated. All the steps described above were performed under dim yellow light.

2.12. Integrated biomarker response

Depending on the results obtained with the biomarkers, the integrated biomarker response (IBR) index was calculated by integrating all the measured biomarker responses into one general stress index to indicate the induction or inhibition of each biomarker and further determine the degree of stress for each concentration at different time points. This method, as described by (Sanchez et al., 2013), was performed as follows: (1) Calculate the mean and standard deviation (SD) values for each treatment and standardise the data for each treatment by the formula $Y = \log(X/X_0)$, where X is the mean value of a biomarker from each treatment, X_0 is the mean value from the control and Y represents the standardised value. (2) Calculate the homogenisation value for each treatment by the formula $Z = (Y - M)/S$, where M is the average of standardised values for all the treatments and S is the SD. The same procedure was also performed for the control group, calculating the homogenisation values (Z_0). (3) Calculate the difference between Z and Z_0 (A), where A is the value of the response of each biomarker at each time point. $A > 0$ represents induction, and $A < 0$ represents inhibition for each biomarker. (4) Finally, the IBR value for each treatment exposed to Roundup at each time point is calculated by

the formula $IBR = \sum_{i=1}^n |Ai|/n$, where n is the quantity of the biomarkers. For the control group, $IBR = 0$. The data for each biomarker were represented in star-plot graphs, indicating the deviation of the investigated biomarkers in relation to the control.

2.13. Statistical analyses

A probit analysis from SPSS V20.0 software was used to estimate the LC_{50} values and their 95% confidence limits. Results were represented as the average values of 3 replicates $\pm$ standard errors. The data were statistically analysed by analysis of variance (ANOVA). A homogeneity test of variances was performed, and a multiple-comparison Duncan test was used to determine significant differences between each group and the control. A P value < 0.05 was considered significant.

3. Results

3.1. LC_{50} of roundup

All the shrimps in the control group survived during the experiment. According to the results of the probit analysis, the 48-h and 96-h LC_{50} values of Roundup on *M. nipponense* were 57.684 mg/L (95% confidence limits: 29.977–141.17) and 11.237 mg/L (95% confidence limits: 2.962–18.111), respectively, and the safe concentration was 0.657 mg/L.

3.2. Effect of roundup on THC

The THC in groups of 1.40, 2.80 and 5.60 mg/L decreased significantly at 24 h ($P < 0.05$), however, it recovered at 48 h except group of 5.60 mg/L. At 96 h, the THC only in concentration group of 5.60 mg/L was significantly lower than control ($P < 0.05$) (Fig. 1).

3.3. Effect of roundup on antioxidant activity

The activities of SOD in serum in each treatment significantly increased at 24 h after exposure to Roundup and sharply decreased at 48 h, where all the treatment groups were significantly lower than the control, except the 0.35 mg/L concentration group. Then, the SOD activities in the 1.40, 2.80 and 5.60 mg/L concentration groups continuously decreased with time and increasing dose, and at 96 h, the lowest level of SOD was observed at a concentration of 5.60 mg/L, at approximately one half of the control (Fig. 2A). A

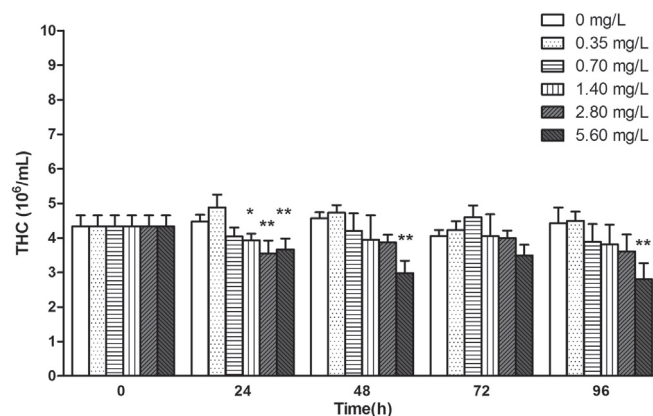


Fig. 1. The THC in *M. nipponense* exposed to Roundup. Note: asterisk over the column represents a significant difference, * represents $P < 0.05$ and ** represents $P < 0.01$ between the experimental group and the control; the same applies to the figures below.

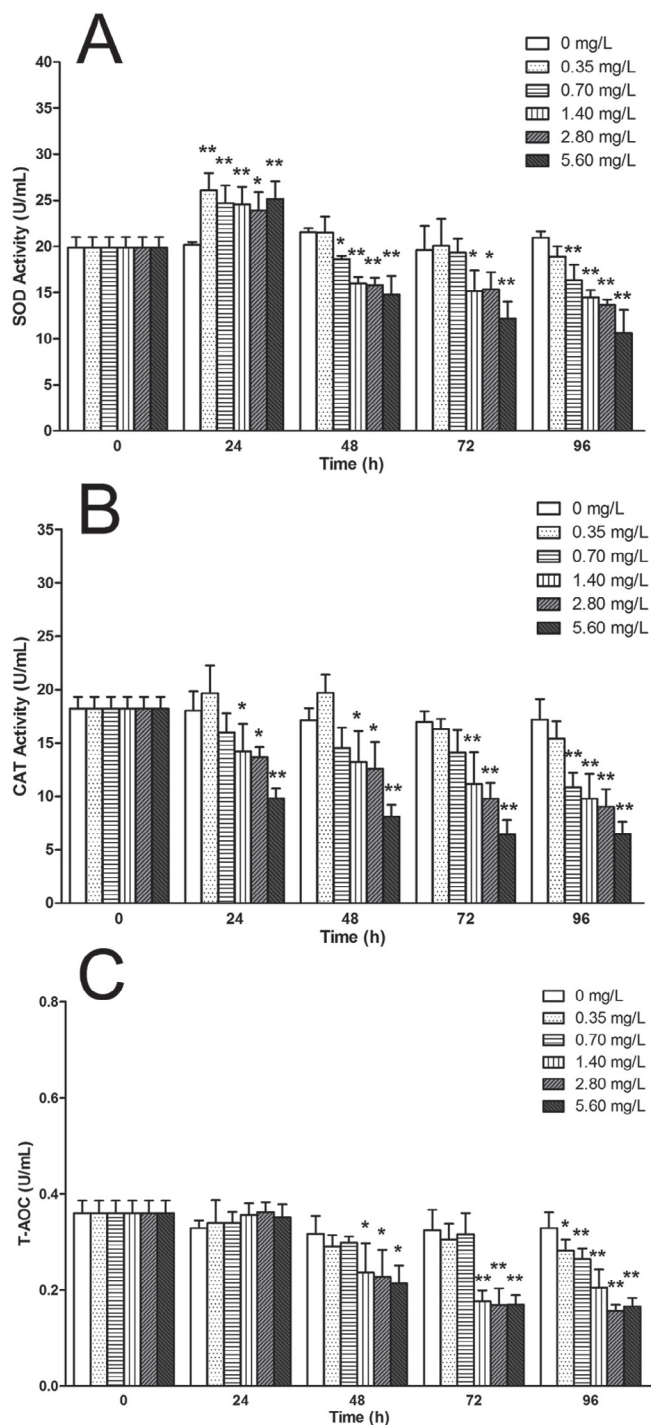


Fig. 2. The antioxidant activity in *M. nipponense* exposed to Roundup. A: SOD activity; B: CAT activity; C: T-AOC.

decrease ($P < 0.05$) in CAT activities with increasing exposure time was observed in all the concentration groups, except the 0.35 mg/L concentration group. There was no difference in the CAT activity between the 0.35 mg/L treatment and the control throughout the experiment ($P > 0.05$). In contrast, a time- and dose-dependent decrease was observed in the other treatment groups, and each group was significantly lower ($P < 0.01$) compared to the control at 96 h (Fig. 2 B). As shown in Fig. 2C, the level of T-AOC slightly increased at 24 h but decreased at 48 h. After 96 h of exposure to

Roundup, the level of T-AOC in all the treatments was significantly lower than that of the control ($P < 0.05$).

3.4. Effect of roundup on parameters of oxidative damage

The MDA levels markedly increased 24 h after treatment of 2.80 and 5.60 mg/L concentrations. The MDA levels peaked at a concentration of 5.60 mg/L at 72 h, with a nearly 2.3-fold increase compared with that of the control. Finally, at 0.70, 1.40, 2.80 and 5.60 mg/L concentrations, the levels of MDA were significantly lower than those of the control ($P < 0.05$) at 96 h (Fig. 3 A). Similarly, the H_2O_2 contents increased in dose- and time-dependent manners in each treatment, except for the 1.40 mg/L treatment, which slightly decreased at 96 h. The highest increase was detected in the 5.60 mg/L concentration group, which was nearly twice that of the control at 96 h (Fig. 3 B). The protein carbonyl content was also increased in the high concentration groups, and after treatment at 5.60 mg/L, the protein carbonyl content peaked, with a 1.6-fold higher level than that of the control at 72 h (Fig. 3C).

3.5. Effect of roundup on acetylcholinesterase activity

The acetylcholinesterase (AChE) activity in serum was significantly diminished ($P < 0.05$) at 24 h after exposure to Roundup in all the treatment groups, except at 0.30 mg/L. With the progress of the experiment, the level of AChE in each concentration continued to decrease, and a dose-dependent response was detected at each time point (Fig. 4).

3.6. Effect of roundup on enzyme kinetics

As shown in Table 1, the K_m values of SOD, CAT and AChE were 0.153 mM, 18.144 mM and 0.624 μ M, respectively. The IC_{50} for AChE was 6.779 mg/L, with $R^2 = 0.9844$, and 1034 mg/L for SOD and 569 mg/L for CAT, with $R^2 = 0.8263$ and 0.7652. It indicated that SOD and CAT may not be affected by Roundup notably but has a marked impact on AChE activity. Fig. 5 showed the relations between enzyme concentrations and residual activities which presented approximate parallel lines among different concentrations of Roundup, and this indicated the inhibitory type of Roundup to AChE as irreversible inhibition.

3.7. Effect of roundup on MN frequency

As shown in Fig. 6, the MN frequency in circulating haemocytes from exposed shrimp varied during the experiment. An obvious increase in the number of MN was observed in specimens at 24 h after exposure to Roundup for the 2.80 mg/L and 5.60 mg/L concentration, and a dose- and time-dependent response in MN frequency was observed throughout the exposure. The MN values at 5.60 mg/L reached the highest level at 96 h, with values approximately 1.3-fold higher than those of the control ($P < 0.01$). The MN frequency in the 1.40, 2.80 and 5.60 mg/L treatments was significantly higher ($P < 0.05$) than that of the control at 96 h, indicating that Roundup at concentrations above (and including) 1.40 mg/L could dramatically induce the occurrence of MN in haemocytes.

3.8. Effect of roundup on DNA damage in haemocytes

Several types of comets were observed in each treatment as early as 24 h, and the level of DNA damage was dependent on the length of the tail and the percentage of DNA in the tail. In the control group, most of the nuclei were circular without any tails, whereas in the 1.40, 2.80 and 5.60 mg/L groups, comets were observed with long tails (Fig. 7). With increasing dose and exposure

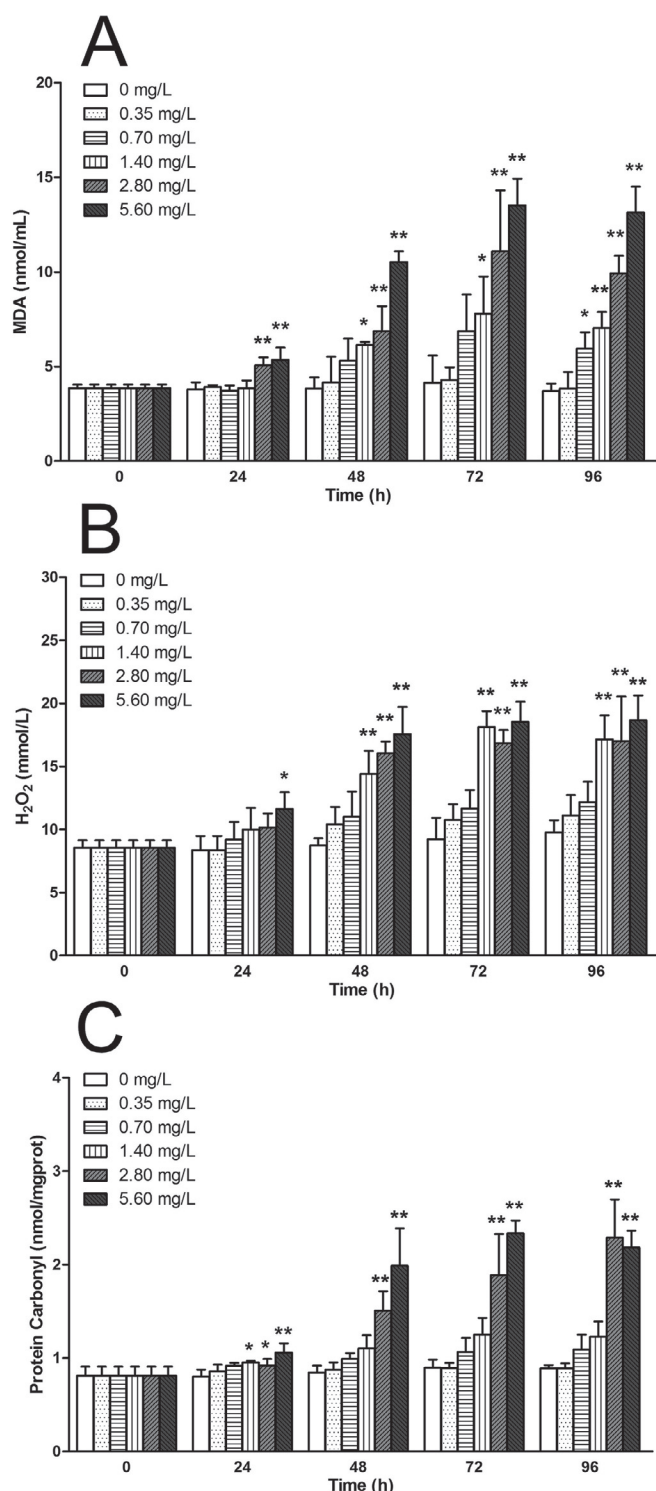


Fig. 3. The oxidative damage parameters in *M. nipponense* exposed to Roundup. A: levels of MDA; B: levels of H_2O_2 ; C: levels of protein carbonyl.

progress, more obvious comet tails were observed in the microscopic fields. Fig. 8 shows an increase in the comet ratio and %DNA in the tail in each treatment from 24 h to 96 h. Both the comet ratio and %DNA in the tails exhibited clear time- and dose-dependent responses during exposure. After 96 h of exposure, all the treatments, except 0.30 mg/L, were significantly higher than the control. In the 5.60 mg/L treatment group, the comet ratio and %DNA in the

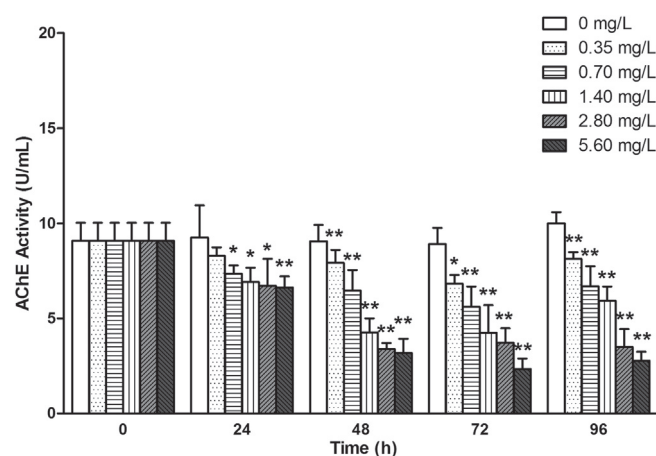


Fig. 4. The acetylcholinesterase activity in *M. nipponense* exposed to Roundup.

Table 1

The inhibition kinetic parameters in SOD, CAT and AChE.

Parameters	SOD	CAT	AChE
K_m	0.153 mM	18.144 mM	0.624 μ M
IC_{50}	1034 mg/L	569 mg/L	6.779 mg/L
R^2	0.8263	0.7652	0.9844

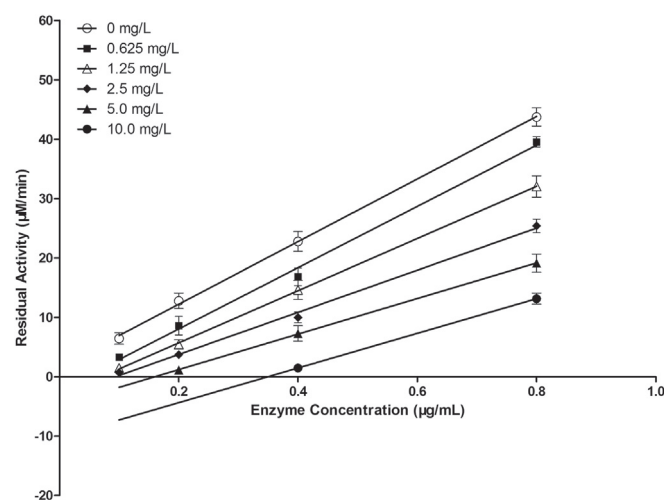


Fig. 5. Linear plot of AChE activity at different Roundup concentrations.

tail increased, with approximately 5.5- and 4.4-fold higher levels than the control.

3.9. Integrated biomarker response

The star plots of the transformed data of all the biomarkers in each treatment group are presented in Fig. 9, and each axis of the star plot represents a biomarker. At 24 h, the areas of SOD, T-AOC, MN, CR, %DNA, H_2O_2 , MDA and PC were more than zero, which indicated the induction of each biomarker at this time point, while the activities of CAT and AChE were inhibited (Fig. 9 A). However, from 48 h to 96 h, all the biomarkers of antioxidants, including T-AOC, SOD, CAT and AChE, in each treatment were inhibited, and all the oxidative stress and genotoxic biomarkers were induced (Fig. 9 B, C, D). These results demonstrated the occurrence of oxidative and genotoxic damage during the exposure of Roundup, even when

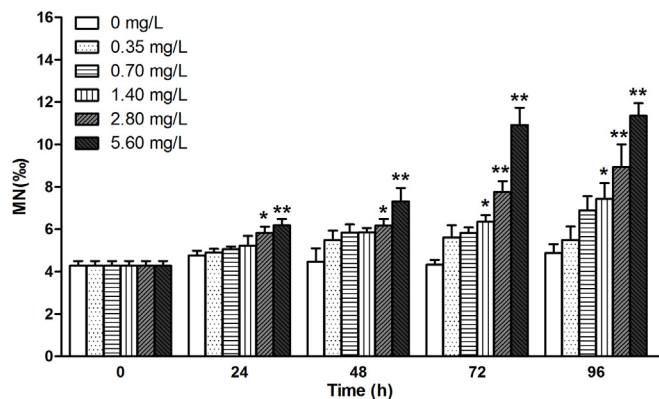


Fig. 6. The MN frequency in haemocytes from *M. nipponense* during exposure to Roundup.

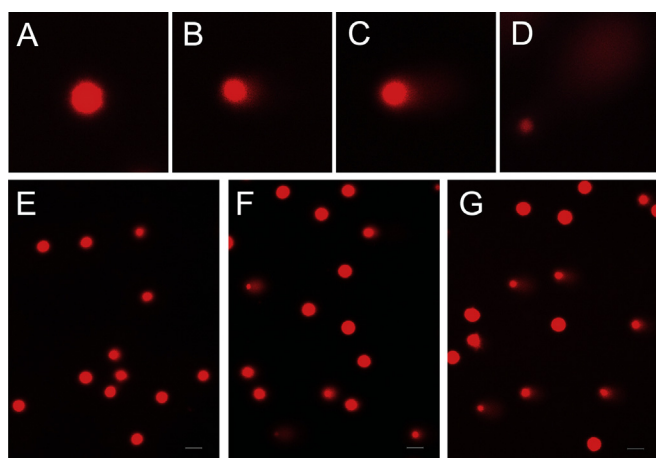


Fig. 7. Different levels of DNA damage in haemocytes from *M. nipponense* after 24 h of exposure to Roundup. A: control without tail, B: comet with tiny tail, C: comet with long tail, D: comet with tiny head and large tail, E: control, F: at the concentration of 1.40 mg/L, G: at the concentration of 5.60 mg/L. Scale bar = 10 μ m.

receiving the lowest dose of 0.35 mg/L. In addition, there was a dose-dependent response for each biomarker from 48 h to 96 h.

The IBR index in Table 2 shows a slight increase at all the concentrations, except 0.35 mg/L. In contrast, the IBR index at 0.35 mg/L gradually decreased, and a dose-dependent response was also observed at each time point. In addition, the IBR values and Roundup concentrations showed an obvious positive correlation at each time point, with $R^2 = 0.923, 0.891, 0.873$ and 0.847 . These results indicate that during the 96-h exposure, concentrations higher than 0.35 mg/L of Roundup could cause oxidative stress and unrecoverable genotoxic effects.

4. Discussion

Glyphosate and its commercial formulation has a prominent toxic effect on fresh water organisms, particularly fish and crustaceans (Mensah et al., 2011). Interestingly, numerous studies have shown that the 96-h LC_{50} value of Roundup dramatically varied depending on the species, life stage and environmental condition (Jiraungkoorskul et al., 2002). For example, the values were 2.45 mg/L for the neonate of *Caridina nilotica* but 27.76 mg/L for the adult (Mensah et al., 2011), 13.69 mg/L for the neotropical fish *Prochilodus lineatus* (Langiano and Martinez, 2008) and 42 mg/L for the cold-water species, *Salmo salar* (Servizi et al., 1987). In addition,

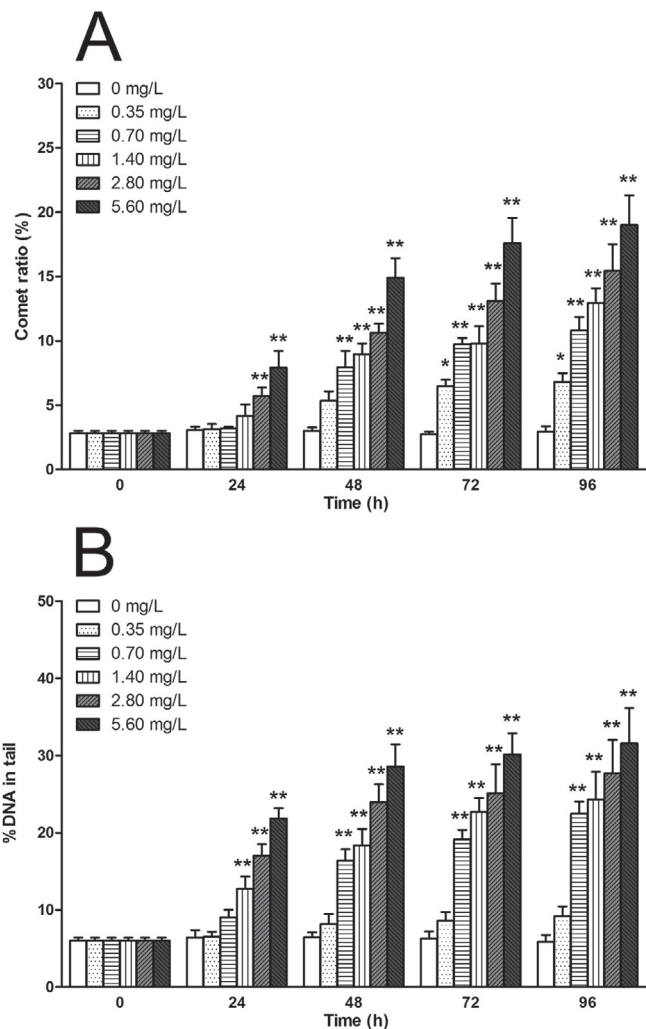


Fig. 8. DNA damage in haemocytes from *M. nipponense* when exposed to Roundup. A: The comet ratio in haemocytes observed under a fluorescence microscope; B: The percentage of DNA in the tail calculated by CASP software.

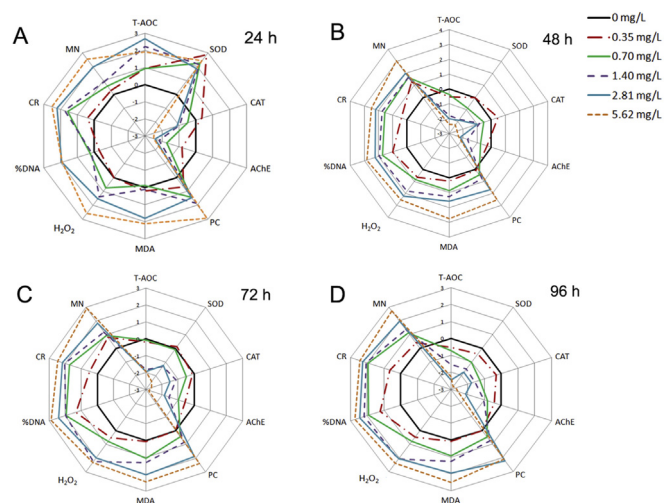


Fig. 9. Star plots for each biomarker from *M. nipponense* when exposed to Roundup. A: 24 h, B: 48 h, C: 72 h, D: 96 h; CR: Comet ratio, PC: Protein carbonyl.

Table 2The IBR index from biomarkers of *M. nipponense* for each treatment.

IBR	24 h	48 h	72 h	96 h
0.35 mg/L	0.68	0.50	0.42	0.45
0.70 mg/L	1.01	0.97	0.85	1.11
1.40 mg/L	1.37	1.54	1.54	1.54
2.80 mg/L	1.90	1.91	1.88	2.10
5.60 mg/L	2.39	2.57	2.52	2.53
Linear (R^2)	0.923	0.891	0.873	0.847

(Osterberg et al., 2012) showed that the 24-h LC₅₀ value of Roundup in juvenile blue crabs reached 316 mg/L. These results suggest the difference of sensitivity in crustaceans to glyphosate-based herbicides due to a variety of behaviours and habitats. Therefore, there is a pressing need to investigate the LC₅₀ of Roundup in important commercial species in herbicide-treated areas. In the present study, the 96-h LC₅₀ value of Roundup in adult *M. nipponense* was 57.684 mg/L, which is much lower than that of 97.89 mg/L in *Eriocheir sinensis*, as reported in a previous study (Hong et al., 2017). This result indicated that the safe concentration of Roundup for *E. sinensis* (4.4 mg/L) could be lethal for *M. nipponense*. The tested sublethal concentrations (from 0.35 mg/L to 5.60 mg/L) could be environmentally relevant, considering that concentrations of glyphosate measured in different water bodies were within the range of sublethal concentrations tested in this study, such as 0.328 mg/L in USA (Battaglin et al., 2009) and between 0.36 and 2.16 mg/L in Brazil (Rodrigues and Almeida, 2005); the residue in the aquatic water of a commercial pond was up to 0.765 mg/L one day after spraying (Zhu et al., 2000). Although the concentration in this test exceeded the freshwater aquatic life standard of 0.065 mg/L and the 0.7 mg/L maximum-contaminated level reported by the Environmental Protection Agency (EPA) (Nwani et al., 2013), the concentration in the aquatic ecosystem may be relatively higher for repeated use in practice in China. This finding indicated the relevance of the tested concentrations for usage guidance.

In the present study, a set of oxidative stress biomarkers of *M. nipponense* were determined to understand the potential oxidative and genotoxic effects of Roundup. Roundup induces oxidative stress in aquatic organisms, such as gold fish, *Carassius auratus* (Lushchak et al., 2001), *Prochilodus lineatus* (Modesto and Martinez, 2010), and *Anabas testudineus* (Samanta et al., 2014). However, most studies have focused on teleostean fishes, and there are few studies concerning crustaceans. Crustaceans have been reported as organisms sensitive to organophosphate pesticides, which could adversely affect their health and survival at relatively low pesticide concentrations (Kumar et al., 2010; Mensah et al., 2011). The exposure of aquatic organisms, including crustaceans, to Roundup may disequilibrate the production and elimination of ROS, leading to a decrease in the efficiency of antioxidant defence or even oxidative damage. SOD and CAT are major enzymes in the first line of defence against oxidative damage by converting the superoxide anion radical to hydrogen and the hydrogen peroxide to water and molecular oxygen, respectively. In this study, the SOD activity in the serum of all the treatments initially increased then dramatically decreased at 48 h and finally decreased in a dose-dependent manner. These results are consistent with a previous study, which indicated that sublethal concentrations of glyphosate can induce the synthesis of SOD, but with the aggravation of oxidative damage, SOD activity is remarkably inhibited (Hong et al., 2017). However, CAT activity showed a dose- and time-dependent decrease throughout the experiment after exposure to Roundup above concentrations of 0.35 mg/L. Similar results have previously been reported (Nwani et al., 2013), demonstrating an evident time-dependent decrease in CAT activity in the liver of *Channa punctatus*

during a 35-day exposure to Roundup. Notably, no alteration in hepatic CAT activity was observed in *Prochilodus lineatus* (Modesto and Martinez, 2010), whereas an increase in hepatic CAT activity was observed in the livers of gold fish when both fish types were exposed to Roundup for 96 h. This contradiction may be due to the tissue-specific differences in antioxidant enzyme activity but, more persuasively, the species-specific differences in antioxidation mechanisms. Nevertheless, because of the variation in CAT activity during Roundup exposure, it is not recommended to use CAT activity as a sensitive and exclusive biomarker in the assessment of the toxicity of herbicides. The level of T-AOC reflect the overall ability of antioxidation in a tissues and is considered a useful tool for detecting oxidative stress in both bodily fluids and tissues (Weifen et al., 2012). In the present study, the T-AOC remarkably decreased, particularly at the 5.60 mg/L concentration. Notably, there was no alteration at the 0.35 and 0.70 mg/L concentrations until 72 h, while the other treatment concentrations resulted in significant decreases. This finding indicated that long-term exposure to Roundup in *M. nipponense* would inhibit antioxidation and that higher concentrations will show worse effects. In addition, a significant decrease was observed in THC at concentration of 5.60 mg/L throughout the experiment. Similar results were reported in the study about *Eriocheir sinensis* when exposed to same herbicide that glyphosate could induce sharp decrease in THC even at concentration of 4.4 mg/L (Hong et al., 2017). Haemocytes play important roles in immune and antioxidative systems and could be easily affected by extrinsic factors such as pesticides (Pimpão et al., 2007). The results suggest that the hypocytosis or hematopoiesis inhibition may one of the reasons to the imbalance of antioxidative system from *M. nipponense* when exposed to Roundup, especially at high concentration (5.60 mg/L).

Meanwhile, three main indicators of oxidative stress, MDA, H₂O₂ and protein carbonyl, were measured to evaluate the potential oxidative damage induced by Roundup exposure. During the test, the occurrence of lipid peroxidation (LPO) was recognised by the increase in MDA levels in serum in dose- and time-dependent manners. The results were in accordance with studies from (Nwani et al., 2013), who showed that there was also a dose- and time-dependent increase in lipid peroxidation levels in the blood of *Channa punctatus* during chronic exposure at concentrations from 3.25 to 6.51 mg/L. Studies of oxidative stress in the livers of *Prochilodus lineatus* caused by Roundup (Modesto and Martinez, 2010) also indicated a significant increase at 24 h and 96 h after exposure to Roundup at a concentration of 10 mg/L. However, there was no significant difference between the 0.35-mg/L concentration group and the control throughout the study. This finding suggests that a relatively low concentration of Roundup may not induce LPO in tissues. Previous studies (Bertrand et al., 2016) have shown that another OP pesticide, chlorpyrifos, induced thiobarbituric acid reactive species (TBARS), indicating significant LPO at a concentration of 94.5 ng/L, while no significant difference was observed above 31.5 ng/L, compared to the control in the shrimp *Palaeomonetes argentinus*. Additionally, the TBARS detected in liver of a hybrid fish *Pseudoplatystoma* sp was significantly increased after exposure to 7.5 mg/L of Roundup, while no alteration was observed at concentrations of 2.25 mg/L and 4.5 mg/L. To some extent, these results reinforced our hypotheses. Similar results were obtained for the level of H₂O₂, which significantly increased at concentrations above 0.70 mg/L. In addition, acute Roundup exposure caused protein oxidation by increasing protein carbonyl at 2.80 mg/L and 5.60 mg/L. This effect is consistent with the decrease in SOD and CAT at the same concentrations during the entire experiment. Consistent with the results of the present study, a previous study (Sinhonin et al., 2014) also observed an increase in protein carbonyl

in the liver of fish with a decrease in SOD and CAT when exposed to the same herbicide. Due to the non-reversible carbonyl formation, the conformational changes in protein structure and carbonyl formation could be causes of the loss of activity of antioxidant enzymes, such as SOD and CAT. The present study provides evidence of this viewpoint. However, the protein carbonyl level in the present study was much lower compared to the results from Sinhorin et al. This difference may be because the liver is the main site of protein carbonyl production. Because of the convenient detection procedure, the measurement of protein carbonyl in serum may be a more sensitive and efficient biomarker in the assessment of herbicide-induced oxidative stress. Remarkably, the increase in protein carbonyl level, similar to the H_2O_2 and MDA levels, was accompanied with a decrease in SOD, CAT and T-AOC activity from 48 h to 96 h, suggesting that the inhibition of antioxidant systems through Roundup exposure could lead to a decline in antioxidant defence and oxidative damage under this circumstance.

The primary effect of OPs in invertebrates is the inhibition of the enzyme AChE, which is responsible for terminating the nerve impulse transmission (Lushchak, 2011). The accumulation of acetylcholine may affect the fleeing and reproductive behaviours of fish due to a reduction in enzyme activity (Bretaud et al., 2000). In the present study, the AChE activity in the serum of *M. nipponense* was detected as a biomarker of herbicide exposure for the first time. The reduction of AChE activity was observed as significantly lower than the control at 48 h in each treatment. Compared to the other parameters, the results indicated that AChE is more sensitive to the exposure of Roundup and suffers oxidative stress at lower exposure concentrations than proteins and antioxidant enzymes. AChE is a sensitive biomarker of crustaceans exposed to pesticides (Tu et al., 2009), and the brain and muscle are the tissues most commonly used in assays (Sancho et al., 2000). However, the results of the present study indicated that AChE in serum could also be a sensitive biomarker in OP exposure, and Roundup could induce the prevention of nerve conduction by the inhibition of AChE activity. Moreover, the decrease in AChE in serum is related to the hepatocyte damage of humans in OPs poisoning (Huang et al., 2009). The present work suggests the potential oxidative injury induced by Roundup exposure. In addition, the data from kinetics assay showed that Roundup has a prominent inhibitory effect on AChE activity, but might not affect SOD and CAT activity directly due to the high IC_{50} and low R^2 values. We suggested that Roundup probably decline the activity of SOD and CAT in tissues from *M. nipponensis* by excess production of peroxide, and more likely, the cellular damage. On the other hand, the linear plot of enzyme concentration against activities of AChE at different concentration of Roundup indicates the irreversible inhibition of Roundup to AChE.

The MN test was developed as a simple and practical in vivo cytogenetic screening method for mutagens (Tsuchimoto and Matter, 1979) and has been widely used in studies for the evaluation of genotoxic effects caused by different xenobiotics on aquatic animals (Bolognesi and Hayashi, 2011; Schmid, 1976). However, the MN test applied in studies of economic decapods, crabs and shrimps has not been reported. In the present study, MN was observed in all treatment groups, and there was a dose- and time-dependent increase in concentrations of 1.40, 2.80 and 5.60 mg/L during the exposure. These results are consistent with those of previous studies, demonstrating that both pure glyphosate and glyphosate-based products could induce MN in vivo and in vitro (Cavalcante et al., 2008; Mañas et al., 2009; Williams et al., 2000). However, no significant alteration was observed in the concentration groups of 0.35 and 0.70 mg/L. Several reports involving fishes with *Tilapia rendalli*, *Carassius auratus*, and *Cnesterodon*

decemmaculatus exposure to Roundup or other glyphosate-based herbicides showed the lack of MN induction within 48 h. These authors (Cavalcante et al., 2008; Mañas et al., 2009; Williams et al., 2000) suggested that this effect could be due to a stimulated erythro-cateretic function with a slow replacement of erythrocytes and their delayed appearance in peripheral blood. The present results may provide evidence for this suggestion. However, more evidence, for instance, the total haemocyte counts and different haemocytes counts in herbicide exposure, are needed to further demonstrate this finding.

In previous studies, DNA damage from aquatic animals such as fishes and bivalves was assessed by comet assays after exposure to genotoxicants (Abd-Allah et al., 1999; Deventer, 1996; Wilson et al., 1998). However, there are few reports related to macrocrustaceans. In the present study, we conducted the comet assay in haemocytes successfully to evaluate the genotoxicity of Roundup on *M. nipponense*. DNA damage was detected at 24 h with concentrations of 2.80 mg/L and 5.60 mg/L, with a comet ratio and %DNA in the tail that were significantly higher than that of the control. The dose- and time-dependent increases in both parameters were observed during the exposure and at 96 h, and the comet ratio in each treatment was significantly higher than that of the control. These findings indicated that DNA in the haemocytes was significantly damaged by exposure to Roundup, even below the safe concentration (0.657 mg/L) and the residue in a culture pond after one day of spraying (0.765 mg/L). The results in the present study are consistent with those of a previous study (Nwani et al., 2013) on the blood and gill cells of *Channa punctatus* after exposure to Roundup at concentrations ranging from 3.25 mg/L to 6.51 mg/L and another previous study on the haemocytes of *E. sinensis* exposed to glyphosate from 4.4 mg/L to 98 mg/L (Hong et al., 2017). However, no decline in DNA damage was observed within 96 h of exposure in the present study, while a decrease in DNA damage followed by a time-dependent increase in DNA damage was found at 14 d in the study of (Nwani et al., 2011), which may be due to the repair of damaged DNA. An explanation for this discrepancy may be the lack of long-term observations in this acute toxic test. Nevertheless, the present data showed that the comet ratio and %DNA in the tail could be sensitive and accurate parameters for the detection of DNA damage with low concentrations of Roundup exposure. Additionally, the increase in the comet ratio and %DNA in the tail was accompanied by an increase in MN frequency and MDA, H_2O_2 and protein carbonyl levels as well as a decrease in antioxidant enzyme activity, including SOD, CAT and T-AOC. As a consequence, it is considered that the DNA damage might be the primary cause of oxidative damage, which led to the lipid peroxidation and chromosomal aberration in cells of immunologic tissues and induced the enzyme activity inhibition.

In the present study, the biomarkers of oxidative stress, including SOD, CAT, T-AOC, MDA, H_2O_2 and protein carbonyl, the neurotoxicity AChE, the genotoxicity MN, and the comet ratio and %DNA in the tail were used to calculate the IBR. These biomarkers have been used to analyse the effects of Roundup exposure separately in previous studies (Modesto and Martinez, 2010; Sinhorin et al., 2014; Tu et al., 2009). However, the present study is the first to conduct an integrated detection of the potential occurrence of genetic and oxidative damage after exposure to sublethal concentrations of Roundup. The results showed the dose-dependent induction of oxidative damage and the inhibition of antioxidant activity during exposure. These findings are consistent with those of a study on the shrimp *P. argentinus* exposed to chlorpyrifos, which used the IBR method to assess the toxic effects of xenobiotic responses and the sensitivity of biomarkers (Bertrand et al., 2016). The IBR method has been used for environmental risk assessment

and to evaluate the toxicological effects of different xenobiotics, including pesticides, to aquatic organisms (Sobjak et al., 2017). A good linear relation between IBR values and Roundup concentrations at each time point was observed in the present study. This association reveals the availability and sensitivity of IBR to evaluate the toxicity of Roundup as a bioindicator and suggests that IBR could be of great application value for the quantification of pollution degrees and integrated responses induced by Roundup towards shrimps, such as *M. nipponensis*.

5. Conclusion

In the present study, a significant oxidative stress and genotoxic effect on *M. nipponensis* was detected with exposure to the glyphosate-based herbicide Roundup at environmentally related concentrations, which induced remarkable DNA damage and antioxidative response inhibition, especially the irreversible inhibition in AChE activity. In addition, the IBR method was performed to evaluate the potential stress induced by the herbicide. The time and dose responses preliminarily showed the occurrence of antioxidant inhibition following cellular damage by oxidations that may lead to the irreparable stress within a short period of 96 h, even under sublethal concentrations. This effect could increase the risk of pathogen invasion in a commercial pond, exposed directly or indirectly to pesticides. Thus, it is best not to use Roundup in aquatic systems, particularly in rice-shrimp coculture ponds. Additionally, all the biomarkers except CAT in the present study are considered sensitive for the toxicity assessment of Roundup exposure.

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